

Numerical Methods

By

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Chapter (2): Solution of Algebraic Equations

The solution (values of x) are known as **the roots** of the equation $f(x) = 0$, or **the zeroes** of the function $f(x)$

There are two types of methods to find the roots of algebraic equations.

Direct Methods:

give the exact value of the roots in a finite number of steps.

Direct methods determine all the roots at the same time

Indirect or Iterative Methods:

Indirect or iterative methods are based on the concept of successive approximations. The general procedure is to start with one or more initial approximation to the root and obtain a sequence of iterates (x_k) which in the limit converges to the actual or true solution to the root.

The indirect or iterative methods are further divided into two categories:

- Bracketing methods
- Open methods.

The bracketing methods :

require the limits between which the root lies

The open methods:

require the initial estimation of the solution.

Bisection and False position methods are two known examples of the bracketing methods.

Among the open methods, the Newton-Raphson and the method of successive approximation are most commonly used.

Contents of the Lecture:

1. Bisection Method
2. Method of False Position (Regular Falsi Method)

Bisection Method

This method is also known as **the interval halving method**

If a function $f(x)$ is continuous on the interval $[a, b]$ and $f(a)f(b) < 0$, then at least one root exists in the interval $[a, b]$

Algorithm for the Bisection Method

1- Choose an interval $[a, b]$ such that $f(a)f(b) < 0$

2- Compute the midpoint:

$$c = \frac{a+b}{2}$$

3- Evaluate $f(c)$

4- Select the new interval:

If $f(a)f(c) < 0$, the root lies in $[a, c]$

If $f(c)f(b) < 0$, the root lies in $[c, b]$

5- If $b - c \leq \epsilon$, then accept c as the root and stop

ϵ is the error tolerance

Bisection Method

To find out how many iterations will be necessary, we use the expression

$$n \geq \frac{\log \left(\frac{b-a}{\epsilon} \right)}{\log 2}$$

Example :

Use the Bisection method to find a root of the equation

$$x^3 - x - 2 = 0$$

In the interval $[1, 2]$, correct to three decimal places.

Solution :

Here, $f(x) = x^3 - x - 2 = 0$

$$f(1) = 1^3 - (1) - 2 = -2 < 0$$

$$f(2) = 2^3 - (2) - 2 = 4 > 0$$

Since $f(1) \cdot f(2) < 0$, a root exists between 1 and 2.

Numbers of Iteration:

$$n \geq \frac{\log \left(\frac{b-a}{\epsilon} \right)}{\log 2} = \frac{\log \left(\frac{2-1}{10^{-3}} \right)}{\log 2} \geq 10$$

Iteration 1: $a_1 = 1$, $b_1 = 2$

Midpoint (c_1): $c_1 = \frac{a_1+b_1}{2} = \frac{1+2}{2} = 1.5$

Evaluate $f(c_1)$: $f(1.5) = (1.5)^3 - (1.5) - 2 = -0.125 < 0$

Update Interval: Since $f(1.5)$ is negative and $f(2)$ is positive,
the new interval is **[1.5, 2]**

Iteration 2: $a_2 = 1.5$, $b_2 = 2$

Midpoint (c_2): $c_2 = \frac{a_2+b_2}{2} = \frac{1.5+2}{2} = 1.75$

Evaluate $f(c_2)$: $f(1.75) = (1.75)^3 - (1.75) - 2 = 1.609 > 0$

$$|c_2 - c_1| = |1.75 - 1.5| = 0.25$$

Update Interval: Since $f(1.5)$ is negative and $f(1.75)$ is positive,
the new interval is **[1.5, 1.75]**

Iteration 3: $a_3 = 1.5$, $b_3 = 1.75$

Midpoint (c_3): $c_3 = \frac{a_3+b_3}{2} = \frac{1.5+1.75}{2} = 1.625$

Evaluate $f(c_3)$: $f(1.625) = (1.625)^3 - (1.625) - 2 = 0.666 > 0$

$$|c_3 - c_2| = |1.625 - 1.75| = 0.125$$

Update Interval: Since $f(1.5)$ is negative and $f(1.625)$ is positive, the new interval is **[1.5, 1.625]**

Iteration 4: $a_4 = 1.5$, $b_4 = 1.625$

Midpoint (c_4): $c_4 = \frac{a_4+b_4}{2} = \frac{1.5+1.625}{2} = 1.5625$

Evaluate $f(c_4)$: $f(1.5625) = (1.5625)^3 - (1.5625) - 2 = 0.252 > 0$

$$|c_4 - c_3| = |1.5625 - 1.625| = 0.0625$$

Update Interval: Since $f(1.5)$ is negative and $f(1.5625)$ is positive, the new interval is **[1.5, 1.5625]**

Iteration 5: $a_5 = 1.5$, $b_5 = 1.5625$

Midpoint (c_5): $c_5 = \frac{a_5 + b_5}{2} = \frac{1.5 + 1.5625}{2} = 1.53125$

Evaluate $f(c_5)$: $f(1.53125) = (1.53125)^3 - (1.53125) - 2 = 0.059 > 0$

$$|c_5 - c_4| = |1.53125 - 1.5625| = 0.03125$$

Update Interval: Since $f(1.5)$ is negative and $f(1.53125)$ is positive, the new interval is **[1.5, 1.53125]**

Iteration 6: $a_5 = 1.5$, $b_5 = 1.53125$

Midpoint (c_6): $c_6 = \frac{a_6 + b_6}{2} = \frac{1.5 + 1.53125}{2} = 1.5156$

Evaluate $f(c_6)$: $f(1.5156) = (1.5156)^3 - (1.5156) - 2 = -0.034 < 0$

$$|c_6 - c_5| = |1.5156 - 1.53125| = 0.01565$$

Update Interval: Since $f(1.5156)$ is negative and $f(1.53125)$ is positive, the new interval is **[1.5156, 1.53125]**

Iteration 7: $a_6 = 1.5156$, $b_6 = 1.53125$

Midpoint (c_7): $c_7 = \frac{a_6 + b_6}{2} = \frac{1.5156 + 1.53125}{2} = 1.5234$

Evaluate $f(c_7)$: $f(1.5234) = (1.5234)^3 - (1.5234) - 2 = 0.012 > 0$

$$|c_7 - c_6| = |1.5234 - 1.5156| = 0.0078$$

Update Interval: Since $f(1.5156)$ is negative and $f(1.5234)$ is positive, the new interval is **[1.5234, 1.53125]**

Iteration 8: $a_7 = 1.5234$, $b_7 = 1.53125$

Midpoint (c_8): $c_8 = \frac{a_7 + b_7}{2} = \frac{1.5234 + 1.53125}{2} = 1.5195$

Evaluate $f(c_8)$: $f(1.5195) = (1.5195)^3 - (1.5195) - 2 = -0.011 < 0$

$$|c_8 - c_7| = |1.5195 - 1.5234| = 0.0039$$

Update Interval: Since $f(1.5195)$ is negative and $f(1.5234)$ is positive, the new interval is **[1.5195, 1.53125]**

Iteration 9: $a_8 = 1.5195$, $b_8 = 1.5234$

Midpoint (c_9): $c_9 = \frac{a_8 + b_8}{2} = \frac{1.5195 + 1.5234}{2} = 1.5215$

Evaluate $f(c_9)$: $f(1.5215) = (1.5215)^3 - (1.5215) - 2 = 0.0006 > 0$

$$|c_9 - c_8| = |1.5215 - 1.5195| = 0.0019$$

Update Interval: Since $f(1.5195)$ is negative and $f(1.5215)$ is positive, the new interval is **[1.5195, 1.5215]**

Iteration 10: $a_9 = 1.5195$, $b_9 = 1.5215$

Midpoint (c_{10}): $c_{10} = \frac{a_9 + b_9}{2} = \frac{1.5195 + 1.5215}{2} = 1.5205$

Evaluate $f(c_{10})$: $f(1.5205) = (1.5205)^3 - (1.5205) - 2 = -0.005$

$$|c_{10} - c_9| = |1.5205 - 1.5215| = 0.001$$

Hence the root is **1.5205** accurate to three decimal places.

Example :

Use the Bisection method to find a root of the equation

$$e^x - 3x = 0$$

In the interval $[1.5, 1.6]$, correct to two decimal places.

Solution :

Here, $f(x) = e^x - 3x = 0$

$$f(1.5) = e^{1.5} - 3(1.5) = -0.01831 < 0$$

$$f(1.6) = e^{1.6} - 3(1.6) = 0.15303 > 0$$

Since $f(1.5) \cdot f(1.6) < 0$, a root exists between 1.5 and 1.6.

Numbers of Iteration:

$$n \geq \frac{\log\left(\frac{b-a}{\epsilon}\right)}{\log 2} = \frac{\log\left(\frac{1.6-1.5}{10^{-2}}\right)}{\log 2} \geq 4$$

Iteration 1: $a_1 = 1.5$, $b_1 = 1.6$

Midpoint (c_1): $c_1 = \frac{a_1+b_1}{2} = \frac{1.5+1.6}{2} = 1.55$

Evaluate $f(c_1)$: $f(1.55) = e^{1.55} - 3(1.55) = 0.0615 > 0$

Update Interval: Since $f(1.5)$ is negative and $f(1.55)$ is positive, the new interval is **[1.5, 1.55]**

Iteration 2: $a_2 = 1.5$, $b_2 = 1.55$

Midpoint (c_2): $c_2 = \frac{a_2+b_2}{2} = \frac{1.5+1.55}{2} = 1.525$

Evaluate $f(c_2)$: $f(1.525) = e^{1.525} - 3(1.525) = 0.0197 > 0$

$$|c_2 - c_1| = |1.525 - 1.55| = 0.025$$

Update Interval: Since $f(1.5)$ is negative and $f(1.525)$ is positive, the new interval is **[1.5, 1.525]**

Iteration 3: $a_3 = 1.5$, $b_3 = 1.525$

Midpoint (c_3): $c_3 = \frac{a_3+b_3}{2} = \frac{1.5+1.525}{2} = 1.5125$

Evaluate $f(c_3)$: $f(1.5125) = e^{1.5125} - 3(1.5125) = 0.0005 > 0$

$$|c_3 - c_2| = |1.5125 - 1.525| = 0.0125$$

Update Interval: Since $f(1.5)$ is negative and $f(1.5125)$ is positive, the new interval is **[1.5, 1.5125]**

Iteration 4: $a_4 = 1.5$, $b_4 = 1.5125$

Midpoint (c_4): $c_4 = \frac{a_4+b_4}{2} = \frac{1.5+1.5125}{2} = 1.50625$

Evaluate $f(c_4)$: $f(1.50625) = e^{1.50625} - 3(1.50625) = -0.00895 < 0$

$$|c_4 - c_3| = |1.50625 - 1.5125| = 0.00625$$

Hence the root is **1.50625 $\approx$ 1.51** accurate to two decimal places.

Example :

Determine the largest root of $f(x) = x^6 - x - 1 = 0$ in the interval $[1,2]$ accurate to within $\epsilon = 0.001$. Use [the Bisection method](#)

Solution :

Try by yourself

False Position Method

The False Position Method is a numerical technique used to find a root of a nonlinear equation

$$f(x) = 0$$

It is similar to the Bisection Method, but instead of taking the midpoint, it uses a secant line to approximate the root.

The equation of a straight line that connects points $(b, f(b))$ to point $(a, f(a))$ is given by:

$$y = \frac{f(b) - f(a)}{b - a} (x - b) + f(b) \quad (1)$$

The points x_s where the line intersects the x -axis is determined by substituting $y = 0$ in Eq.(1) and solving the equation for x .

$$x_s = \frac{af(b) - bf(a)}{f(b) - f(a)} \quad (2)$$

False Position Method

Algorithm for the False Position Method:

- 1- Define the first interval (a, b) such that solution exists between them. Check $f(a) f(b) < 0$.
2. Compute the first estimate of the numerical solution x_{s_1} using Eq.(2).
3. Find out whether the actual solution is between a and x_{s_1} or between x_{s_1} and b .
 - ❑ If $f(a) f(x_{s_1}) < 0$, the solution is between a and x_{s_1} .
 - ❑ If $f(b) f(x_{s_1}) < 0$, the solution is between x_{s_1} and b .
4. Select the subinterval that contains the solution (a to x_{s_1} , or x_{s_1} to b) is the new interval (a, b) and go back to step 2.

Step 2 through 4 are repeated until a specified tolerance or error bound is attained.

Example :

Using the False Position method, find a root of the function

$$f(x) = e^x - 3x^2$$

correct to 5 decimal places in the interval (0.5, 1).

Solution :

Here, $f(x) = e^x - 3x^2 = 0$

$$f(0.5) = e^{0.5} - 3(0.5)^2 = 0.89872 > 0$$

$$f(1) = e^1 - 3(1)^2 = -0.28172 < 0$$

Since $f(0.5) \cdot f(1) < 0$, a root exists between 0.5 and 1.

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Iteration 1: $a = 0.5, b = 1$

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0.5(-0.28172) - 1(0.89872)}{-0.28172 - 0.89872} = 0.88067$$



Evaluate $f(x_1)$: $f(0.88067) = e^{0.88067} - 3(0.88067)^2 = 0.08577 > 0$

Update Interval: Since $f(1)$ is negative and $f(0.88067)$ is positive, the new interval is **[0.88067, 1]**

Iteration 2: $a = 0.88067, b = 1$

$$x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(0.88067)(-0.28172) - 1(0.08577)}{-0.28172 - 0.08577} = 0.90852$$

Evaluate $f(x_2)$: $f(0.90852) = e^{0.90852} - 3(0.90852)^2 = 0.0441 > 0$

$$|x_2 - x_1| = |0.90852 - 0.88067| = 0.02785$$

Update Interval: Since $f(1)$ is negative and $f(0.90852)$ is positive, the new interval is **[0.90852, 1]**

Iteration 3: $a = 0.90852, b = 1$

$$x_3 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(0.90852)(-0.28172) - 1(0.0441)}{-0.28172 - 0.0441} = 0.90993$$



Evaluate $f(x_3)$: $f(0.90993) = e^{0.90993} - 3(0.90993)^2 = 0.00022 > 0$

$$|x_3 - x_2| = |0.90993 - 0.90852| = 0.00141$$

Update Interval: Since $f(1)$ is negative and $f(0.90993)$ is positive, the new interval is **[0.90993, 1]**

Iteration 4: $a = 0.90993, b = 1$

$$x_4 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(0.90993)(-0.28172) - 1(0.00022)}{-0.28172 - 0.00022} = 0.9100$$

Evaluate $f(x_4)$: $f(0.9100) = e^{0.9100} - 3(0.9100)^2 = 0.00001 > 0$

$$|x_4 - x_3| = |0.9100 - 0.90993| = 7 \times 10^{-5}$$

The root is **0.91** accurate to five digits.

Example :

Using the False Position method, find a root of the function

$$f(x) = \cos x - 3x + 5 = 0$$

correct to 3 decimal places in the interval $(0, \pi)$

Solution :

Here, $f(x) = \cos x - 3x + 5 = 0$

$$f(0) = \cos(0) - 3(0) + 5 = 6 > 0$$

$$f(\pi) = \cos(\pi) - 3(\pi) + 5 = -3\left(\frac{\pi}{2}\right) + 4 = -5.4248 < 0$$

Since $f(0) \cdot f(\pi) < 0$, a root exists between 0 and π .

Iteration 1: $a = 0, b = \pi$

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0(-5.4248) - (\pi)(6)}{-5.4248 - 6} = 1.6494$$

Evaluate $f(x_1)$: $f(1.6494) = \cos(1.6494) - 3(1.6494) + 5 = 0.0915 > 0$

Update Interval: Since $f(1.6494)$ is positive and $f(\pi)$ is negative, the new interval is **$[1.6494, \pi]$**

Iteration 2: $a = 1.6494, b = \pi$

$$x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(1.6494)(-5.4248) - (\pi)(0.0915)}{-5.4248 - 0.0915} = 1.6718$$

Evaluate $f(x_2)$: $f(1.6718) = \cos(1.6718) - 3(1.6718) + 5 = 0.0126 > 0$

$$|x_2 - x_1| = |1.6718 - 1.6494| = 0.0224$$

Update Interval: Since $f(1.6718)$ is positive and $f(\pi)$ is negative, the new interval is **$[1.6718, \pi]$**

Iteration 3: $a = 1.6718, b = \pi$

$$x_3 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{(1.6718)(-5.4248)-(\pi)(0.0126)}{-5.4248-0.0126} = 1.6748$$

Evaluate $f(x_3)$: $f(1.6748) = \cos(1.6748) - 3(1.6748) + 5 = 0.0024$

$$|x_3 - x_2| = |1.6748 - 1.6718| = 0.003$$

Update Interval: Since $f(1.6748)$ is positive and $f(\pi)$ is negative, the new interval is **$[1.6748, \pi]$**

Iteration 4: $a = 1.6748, b = \pi$

$$x_4 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{(1.6748)(-5.4248)-(\pi)(0.0024)}{-5.4248-0.0024} = 1.6758$$

Evaluate $f(x_3)$: $f(1.6758) = \cos(1.6758) - 3(1.6758) + 5 = 0.0002$

$$|x_3 - x_2| = |1.6758 - 1.6748| = 1 \times 10^{-3}$$

The root is **1.6758** accurate to Three digits.

Example :

Using [the False Position method](#), find a root of the function

$$f(x) = x^4 - 11x + 8 = 0$$

correct to 4 decimal places in the interval (1,2)

Solution :

Try by yourself

**THANK YOU
FOR YOUR
ATTENTION**

